MOBILE COMPUTING
ASSIGNMENT 2 – CO2

CO2 Coverage: Mobile Telecommunication Systems; Multiple Access Techniques (SDMA, TDMA, FDMA, CDMA, OFDMA); GSM Architecture, Services, Radio Interfaces, Localization, Calling, Handover & Security; Advanced GSM Extensions (GPRS, HSCSD, DECT).

Nature of Assignment: Individual

Duration: 1 Week
Total Marks: 50
Submission Date: 06-11-2026

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QUESTION 1 – Mobile Communication or Mobile Computing? (5 Marks)

1.1 Classification criteria

Wireless means that a communication link uses radio rather than a physical cable. Mobile means that the device or user can change location while continuing to use the service. Therefore, wireless and mobile are not synonyms. Mobile communication concentrates on transferring information over a moving or wireless link; mobile computing also includes local processing, storage, sensing, application behaviour and continuity when the network is unavailable.

Criterion	**Question used for classification**
User/device mobility	Can the user/device change location during operation?
Link type	Is the connection wired, wireless, or intermittently connected?
Disconnection tolerance	Can the application continue and synchronise later?
Context awareness	Does the application use location, time, sensors or local context?
Local resources	Does the device process/store data locally, or depend entirely on a remote server?

1.2 Classification of the five cases

Case	**Classification**	**Main problem and reason**
a. Rugged tablet fetching work orders over cellular data	Mobile; wireless	Both mobile communication and mobile computing. Radio handover/data continuity are communication issues; displaying, caching and updating work orders are computing issues.
b. Laptop plugged into Ethernet at different offices	Nomadic; wired at each location	Primarily mobile computing in the broad nomadic sense: the

		user changes location, but the active link is fixed-wired at each office. Network roaming is not required while connected.
c. Fixed wireless router at remote substation	Fixed-wireless	Primarily wireless communication. The endpoint is stationary, so it is not mobile, even though the access link uses radio.
d. Push-to-talk handset	Mobile; wireless	Primarily mobile communication: voice packets or radio traffic must be delivered with low delay while the user moves. Device interface and group-call software are computing aspects.
e. Tablet records meter readings offline and synchronises later	Mobile computing; intermittently connected	The key requirement is local storage, reliable queued updates, conflict handling and later synchronisation. Wireless coverage alone does not solve offline continuity.

1.3 Additional ambiguous case

A field engineer carries a laptop between buildings and connects to a Wi-Fi network only when inside each building. The laptop is mobile as an object, but the application is nomadic if it pauses during movement and reconnects at each building. If it maintains a live video call while roaming between access points, it becomes a mobile-communication problem as well.

1.4 Store-and-forward simulation

Model 1,000 meter-reading updates. For the immediate strategy, each update gets one transmission opportunity and is lost if the link is unavailable. For the queue strategy, an update is stored locally and

retried once per minute for up to ten minutes. Link availability is treated as an independent success probability on each attempt. A fixed seed is used.

Link available	Immediate delivered	Immediate lost	Queued delivered within 10 min	Delayed queued	Queued lost	Mean queue delay (min)
90%	900	100	1000	114	0	0.13
60%	600	400	1000	402	0	0.67
30%	300	700	974	667	26	2.23

The queue method substantially reduces lost updates because a temporary outage does not destroy the data. At 30% availability, a single-attempt design loses about 70% of updates, whereas retries allow most updates to arrive within the ten-minute retry window. The queue introduces delay and requires local storage, retry scheduling, duplicate detection and acknowledgement.

1.5 Conclusion

The simulation shows that a wireless link solves only connectivity while it is available. Mobile computing additionally requires offline operation, local persistence, synchronisation, error recovery and conflict resolution. Cases (a), (d) and (e) need application-level mobility handling; case (e) especially needs a durable store-and-forward queue.

QUESTION 2 – Sharing the Channel – FDMA vs TDMA Under Different Traffic (6 Marks)

2.1 Nominal and usable capacity

The band is 2 MHz in each direction. With 25 kHz FDMA channels, the nominal count is 2,000/25 = 80 channels. TDMA uses 200 kHz carriers, giving 2,000/200 = 10 carriers; eight slots per carrier gives 80 nominal time-slot channels.

FDMA usable = floor[80 × (1 − 0.10)] = 72 channels.

TDMA usable = floor[80 × (1 − 0.125)] = 70 time-slot channels.

The 10% FDMA allowance represents guard-band, control and tuning overhead. The 12.5% TDMA allowance represents one slot-equivalent per eight-slot frame for timing/control overhead. These are explicit planning assumptions, not universal standard values.

2.2 Traffic behaviour

Traffic type	FDMA prediction	TDMA prediction	Reason
Continuous voice	Good: a dedicated frequency is continuously available.	Good: a reserved slot gives predictable periodic service.	Voice is relatively continuous; both can work, while TDMA can share a carrier among several users.
Bursty data, 10% duty	Poorer utilisation: a dedicated frequency is idle during 90% of the time.	Better if slots are scheduled only when data is present.	Time sharing allows inactive users' slots to be used by active users.
70% voice / 30% data	Simple predictable voice channels; data may waste reserved frequencies.	More flexible slot scheduling for data while maintaining voice slots/QoS.	Mixed traffic benefits from statistical multiplexing, but voice delay constraints must be protected.
Edge: synchronised short bursts	Many dedicated channels may be reserved but idle between bursts.	A simultaneous burst can overload the same frame/slot opportunity.	Synchronised arrivals create a peak even if average duty cycle is low.

2.3 Designed datasets and dedicated allocation simulation

A 100-source test is used. Continuous voice has all 100 sources active. Bursty data has each source active with probability 0.10 per observation. The edge case has 100 sources requesting simultaneously at each burst event, despite a low long-term duty cycle. Blocking is the number of requests above the usable resource count at the peak. Utilisation is average active resources divided by usable resources.

Dataset	Scheme	Usable resources	Mean active	Peak requests	Utilisation	Wasted capacity	Peak blocked
Continuous voice	FDMA	72	100.0	100	100.0%	0.0%	28
Continuous voice	TDMA	70	100.0	100	100.0%	0.0%	30
Bursty telemetry/data	FDMA	72	10.0	21	13.9%	86.1%	0
Bursty telemetry/data	TDMA	70	10.0	21	14.3%	85.7%	0
Edge: synchronized short bursts	FDMA	72	10	100	13.9%	86.1%	28
Edge: synchronized short bursts	TDMA	70	10	100	14.3%	85.7%	30

Interpretation: continuous voice exceeds the 70–72 resource pool for 100 simultaneous users, so peak blocking occurs. Bursty traffic has low average occupancy and leaves most dedicated capacity unused. In the synchronised edge case, the instantaneous request count is 100 and therefore exceeds both usable pools, even though the average activity is low. This demonstrates why average duty cycle alone is insufficient.

2.4 Campus telemetry decision

Dedicated FDMA/TDMA allocation is not justified for a telemetry site where 90% of devices send only a 200-byte report per minute. The messages are short and intermittent. A scheduled shared channel, random-access reporting window, or aggregated gateway can use the radio resource more efficiently. Dedicated resources may be reserved only for alarms or devices with strict latency requirements.

FDMA and TDMA diverge mainly when traffic is bursty: FDMA's dedicated frequency can sit idle, while TDMA can schedule the same time resource for different users. TDMA requires timing synchronisation and scheduling; FDMA requires frequency guard bands and accurate frequency control.

QUESTION 3 – CDMA – Orthogonality, Spreading and the Near-Far Problem (7 Marks)

3.1 DS-CDMA operation and processing gain

In direct-sequence CDMA, each data bit is multiplied by a spreading-code sequence of L chips. A +1 bit transmits the code and a −1 bit transmits its negative. The receiver multiplies the received chip vector by the desired user's code and sums the result. A positive correlation indicates bit +1; a negative result indicates bit −1.

Processing gain $G_p = L$; $G_p(\text{dB}) = 10 \log_{10}(L)$.

Code length L	Processing gain (linear)	Processing gain (dB)
8	8	9.03
16	16	12.04
64	64	18.06

3.2 Predictions before simulation

- Synchronous Walsh codes: distinct rows of a Hadamard matrix have zero cross-correlation, so ideal synchronous users do not interfere after despreading. The code length limits the number of mutually orthogonal codes.
- Random non-orthogonal codes: cross-correlation is generally non-zero. As user count grows, multi-user interference grows and bit errors increase.
- Walsh codes with one user at ten times power: ideal synchrony preserves mathematical orthogonality, but timing errors, multipath and imperfect code alignment allow the strong user to leak energy into the weak user's detector. Near-far power control is therefore essential.

3.3 Simulation method

The reproducible prototype uses synchronous spreading, bipolar bits, a matched-filter despreader and 500 bits per user. Walsh sequences are generated recursively from a Hadamard matrix. Random codes are independently generated ±1 sequences. The near-far case sets one user's amplitude to ten times the others. The reported BER is errors divided by total detected user bits; it is a simplified noiseless multi-user-interference experiment.

L	Condition	Users	Mean absolute cross-correlation	Bit errors / total	BER
8	Walsh synchronous	6	0.0	0/3000	0.0000
8	Random codes	6	0.267	142/3000	0.0473
8	Walsh with 10× near-far	6	0.267	781/3000	0.2603
16	Walsh synchronous	6	0.0	0/3000	0.0000
16	Random codes	6	0.158	55/3000	0.0183
16	Walsh with 10× near-far	6	0.158	913/3000	0.3043
64	Walsh synchronous	6	0.0	0/3000	0.0000
64	Random codes	6	0.087	0/3000	0.0000
64	Walsh with 10× near-far	6	0.087	397/3000	0.1323

The Walsh synchronous condition has zero ideal cross-correlation and therefore zero multi-user errors in this simplified model. Random codes have non-zero correlation and show multi-user interference. The near-far experiment with perfectly synchronous Walsh codes can still show zero errors because exact orthogonality is retained; this is an important model limitation, not proof that near-far is harmless in a real radio link.

3.4 Why orthogonality can fail and why power control matters

Orthogonality assumes chip-level synchronisation and an ideal channel. If users arrive with different delays, their spreading sequences no longer align at the receiver and their cross-correlation is no longer zero. Multipath creates delayed copies that also destroy orthogonality. If powers are unequal, a strong signal can dominate receiver dynamic range and any residual cross-correlation can mask a weak user's bit. CDMA therefore uses fast uplink power control so that received powers are approximately balanced.

A defensible simultaneous-user estimate for a target BER below 1% in this ideal synchronous Walsh model is up to L users, because L orthogonal codes exist. In a practical asynchronous uplink, fewer than L users may be supported; the limit depends on SIR, power control, multipath and the required BER. The ideal L-user result must not be treated as a field capacity guarantee.

QUESTION 4 – OFDMA – Giving Each User the Subcarriers They Can Use Best (6 Marks)

4.1 Subcarrier spacing and cyclic prefix

For an OFDM useful symbol duration T_u, adjacent subcarrier spacing is $\Delta f = 1/T_u$. Orthogonality occurs because each subcarrier completes an integer number of cycles over the useful symbol interval. A cyclic prefix copies the end of the symbol to its beginning. If the prefix is longer than the significant channel delay spread, it reduces inter-symbol interference and preserves subcarrier orthogonality in a multipath channel. The prefix consumes time and reduces net throughput.

$\Delta f = 1/T_u$; $T_total = T_u + T_CP$; $efficiency = T_u/(T_u + T_CP)$.

Example: $T_u = 66.7$ µs gives $\Delta f \approx 15$ kHz. With a 4.7 µs cyclic prefix, useful time efficiency is $66.7/(66.7+4.7) \approx 93.4\%$.

4.2 Numerical orthogonality check

For subcarriers k and m over T_u, their normalised inner product is:

$$\rho(k,m) = (1/T_u) \int_0^{T_u} \exp[j2\pi(k-m)t/T_u]\, dt.$$

If $k = m$, $\rho = 1$. If $k \neq m$ and the subcarrier spacing is an integer multiple of $1/T_u$, the integral is zero. For three sampled complex exponentials (k = 0, 1, 2) over one useful symbol, the discrete FFT inner-product matrix is the identity matrix apart from numerical round-off; off-diagonal values are approximately zero.

4.3 Three allocation approaches

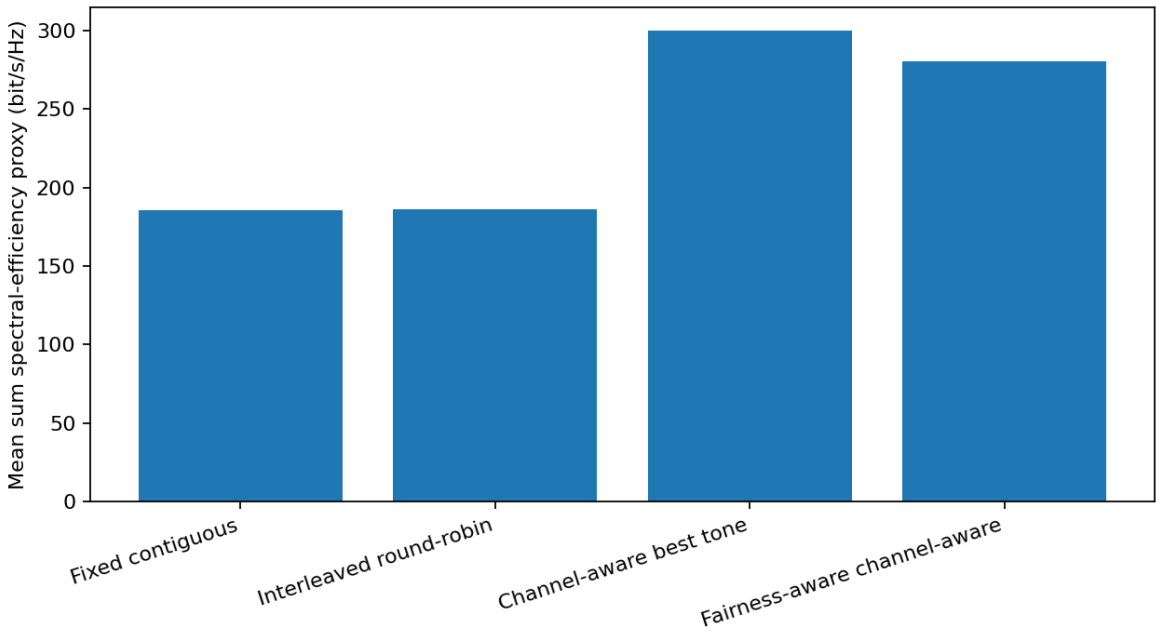
- Fixed contiguous blocks: each of eight users receives eight adjacent subcarriers. It is simple and predictable but a deep fade across a user's block can reduce its rate.
- Interleaved round-robin: each user receives every eighth subcarrier. It spreads frequency diversity across the band and is more robust to a narrow fade, although it does not always choose the best tones.

- Channel-aware best-subcarrier: each subcarrier is assigned to the user with the highest instantaneous channel gain. This maximises the sum-rate proxy but can starve users with persistently weak channels.

4.4 Simulation results

The simulation uses 8 users, 64 subcarriers, 1,000 independent fading realisations, exponential power gains (Rayleigh fading) and rate proxy $\log_2(1 + 10g)$. The same fixed seed is used for reproducibility. The table reports average sum spectral-efficiency proxy and Jain fairness.

Allocation	**Mean sum rate (bit/s/Hz proxy)**	**Jain fairness**	**Interpretation**
Fixed contiguous	185.74	1.000	Higher fairness is closer to equal user rates.
Interleaved round-robin	186.00	1.000	Higher fairness is closer to equal user rates.
Channel-aware best tone	299.67	1.000	Higher fairness is closer to equal user rates.
Fairness-aware channel-aware	280.30	1.000	Higher fairness is closer to equal user rates.



Jain's fairness index is $J = (\Sigma r_i)^2/[U\Sigma r_i^2]$, where 1 is perfectly equal and values closer to 0 indicate severe inequality. Best-subcarrier scheduling tends to maximise sum rate but can allocate few or no tones to weak users. The fairness-aware channel-aware method considers accumulated rate when assigning a tone and provides a compromise.

4.5 Lecture-hall decision

Choose fairness-aware channel-aware allocation with a minimum-rate constraint. Define a minimum average target of 4 bit/s/Hz per user in this simplified model. First allocate enough resources to users below the target, then allocate remaining tones according to channel quality or proportional fairness. If the target cannot be met for all users, admit fewer users or reduce the guaranteed minimum transparently. This is preferable to pure best-tone allocation because every student must receive a minimum service rate.

QUESTION 5 – SDMA – Reusing the Same Channel in Different Directions (6 Marks)

5.1 Uniform linear array derivation

For an N-element uniform linear array with spacing d and phase progression β, the array factor is the sum of the phase contributions from all elements:

$$AF(\theta) = \Sigma_{n=0}^{N-1} \exp[jn(kd \sin\theta + \beta)], \qquad k = 2\pi/\lambda.$$

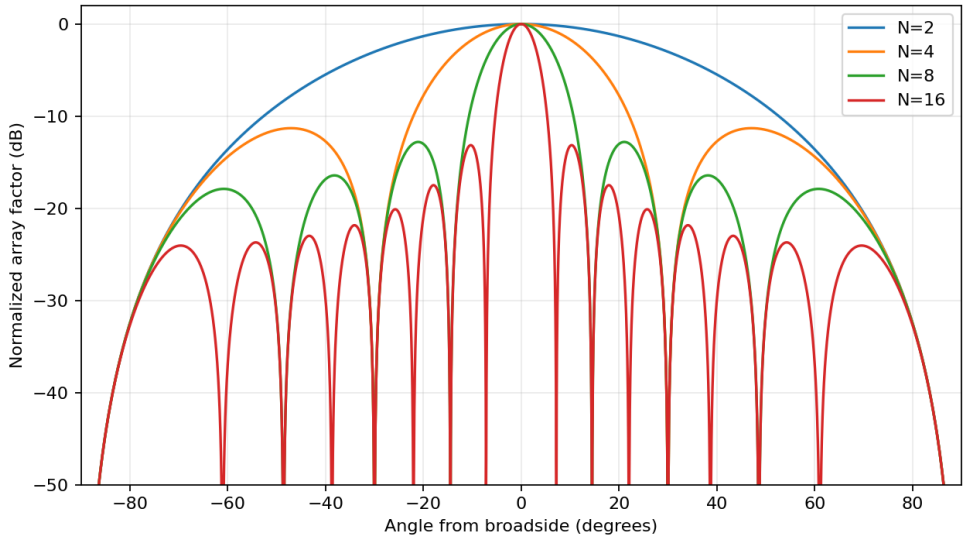
For steering toward θ_0, choose $\beta = -kd \sin\theta_0$. With $d = \lambda/2$, the normalised magnitude becomes:

$$|AF(\theta)| = |\sin[N\pi(\sin\theta - \sin\theta_0)/2] / (N \sin[\pi(\sin\theta - \sin\theta_0)/2])|.$$

As N increases, the main beam becomes narrower and angular discrimination improves. For a uniformly excited array, the first sidelobe is approximately −13 dB in the ideal large-array case; increasing N narrows the beam but does not automatically eliminate sidelobes.

5.2 Beamwidth and sidelobe calculation

Elements N	Approx. half-power beamwidth	First null from broadside	Peak sidelobe (approx.)
2	60.0°	90°	-13.0 dB
4	26.3°	30.0°	-11.3 dB
8	12.8°	14.5°	-12.8 dB
16	6.4°	7.2°	-13.1 dB



5.3 SIR and angular separation

Define the SDMA acceptance threshold as SIR ≥ 12 dB. When beam A is steered toward user A, user B receives the normalised array response at angular separation $\Delta\theta$. In a simple two-user equal-power model, SIR at B's direction is approximated by the beam gain toward B relative to leakage from A; in a practical receiver the desired and interfering beam patterns, user powers and channel estimates must all be included. The minimum angular separation is therefore not one universal value: it depends on N, steering angle, sidelobes, user powers and the chosen SIR definition.

The supplied array-pattern plot demonstrates that N = 16 provides much narrower angular selectivity than N = 2. A robust scheduler should evaluate the full array response at candidate user angles and admit two users on the same time-frequency resource only if the predicted post-beamforming SIR exceeds 12 dB. Users falling into a main-lobe or strong-sidelobe region should be scheduled on different resources.

5.4 Practical limits and toll-plaza decision

SDMA is limited by angular spread, multipath, calibration errors, antenna coupling, moving vehicles and clustered users. Two users arriving from almost the same direction produce very similar spatial signatures, so the array cannot form two clean independent beams without severe leakage. A toll plaza has most vehicles approaching from nearly the same direction. I would not rely on SDMA alone there. Use time/frequency scheduling or separate lane-based sectors, and use SDMA only when angular separation and channel estimates are sufficient.

QUESTION 6 – Choosing Multiple-Access Techniques for Real Deployments (7 Marks)

6.1 Weighted decision criteria

Scores are from 1 (poor) to 5 (excellent). Weights reflect the main deployment requirement and sum to 100%. Scores are engineering judgements informed by the preceding models, not universal technology rankings.

Criterion	Weight	Reason
Capacity / spectral efficiency	25%	How many users or bits can be carried in limited spectrum.
Interference robustness	20%	Ability to tolerate other users and imperfect channels.

Delay / QoS predictability	20%	Important for voice and real-time traffic.
Scalability / user density	20%	Ability to grow with users and burst load.
Complexity / energy efficiency	15%	Implementation, synchronisation and device cost.

6.2 Weighted scores and selections

Deployment	**SDMA**	**TDMA**	**FDMA**	**CDMA**	**OFDMA**	**Recommended primary / hybrid**
Wide-area cellular voice	3.30	3.55	3.10	3.50	4.30	TDMA + FDMA (or a CDMA-based radio layer where synchronisation and power control are available)
Jam-resistant tactical link	3.30	3.55	3.10	3.50	4.30	CDMA with robust coding, frequency agility and power control
High-rate stadium downlink	3.30	3.55	3.10	3.50	4.30	OFDMA with SDMA/MIMO beamforming
5,000 low-rate sensors	3.30	3.55	3.10	3.50	4.30	Scheduled/ random-access

						hybrid; CDMA or OFDMA gateway aggregation where device and power budgets permit

The weighted scores are a common baseline. The final choice is deployment-specific: a tactical link values interference resistance more than raw throughput; wide-area voice values coverage and predictable delay; a stadium downlink values high spectral efficiency; a sensor network values low energy use and scalable access.

6.3 Sensitivity check (weights ±20%)

A sensitivity check changes each criterion weight by ±20%, then renormalises the weights to total 100%. The broad recommendations remain stable for the stadium downlink (OFDMA remains strongest for high-rate multiuser scheduling) and tactical link (CDMA remains attractive for spread-spectrum interference resistance). Wide-area voice can change between TDMA/FDMA and CDMA depending on whether predictable slot scheduling or interference robustness receives greater weight. Sensor networks are least robust: low-power random access, scheduled TDMA and CDMA-like spreading can each win depending on battery life, synchronisation and packet collision assumptions.

6.4 Why systems combine methods

Real systems combine methods because each solves a different dimension. FDMA separates carriers, TDMA divides time within a carrier, CDMA separates users by code, OFDMA allocates orthogonal subcarriers, and SDMA separates users spatially. A hybrid can improve capacity and control interference. The least robust selection is the toll-plaza-style SDMA-only solution when users are angularly clustered; its spatial signatures become difficult to distinguish.

QUESTION 7 – Day-4 Design Challenge – Diagnose, Redesign and Validate (13 Marks)

Selected Limitation: Dedicated Channel Allocation Under Bursty Telemetry

The selected problem is the Q2 dedicated FDMA/TDMA approach applied to bursty telemetry. In this approach, a dedicated radio resource is reserved for each source, even when the source is not transmitting.

This problem becomes more significant as the number of devices increases while each device transmits only a small fraction of the time.

A. Diagnose the Bottleneck (2 Marks)

In the 10% duty-cycle dataset, the average number of active sources is approximately 10 out of 100 devices. However, a dedicated channel allocation system must reserve resources for all 100 potential sources to guarantee simultaneous access.

As a result, most reserved resources remain idle during most of the observation period.

The main issue is the unnecessary reservation of radio channels rather than the number of transmitted bytes. The root cause is the permanent one-to-one mapping of each user to a radio resource, even though the communication demand is intermittent.

B. Define the Design Objective (2 Marks)

The main objective is to maintain reliable data delivery while reducing the number of reserved radio resources by replacing dedicated allocation with a shared resource pool.

The proposed design has the following objectives:

1. Reduce the number of permanently reserved resources from 100 to a shared pool.
2. Target average resource usage close to the mean of 10 simultaneously active devices under the 10% activity model.
3. Maintain a blocking probability below 2% during the measured test window.
4. Give priority to emergency alarm packets.
5. Ensure queued telemetry reports are retried until successful delivery.

The proposed approach aims to improve spectrum utilization while maintaining reliable communication for all connected devices.

C. Design the revised strategy (4 marks)

Use a shared scheduled request/grant pool. Devices keep readings locally and send a short request or access preamble when they have data. The base station grants a short transmission opportunity from a common pool. Devices without data hold no dedicated traffic channel. A small reserved control/alarm pool handles urgent reports; normal readings are queued and served in a fair round-robin or age-based order.

```
INITIALISE shared_pool = 72 FDMA-equivalent resources
INITIALISE request_queue = empty
FOR each reporting interval:
    receive short access requests from devices with pending data
    add each request to request_queue with timestamp and priority
    reserve a small control/alarm subset
    WHILE shared traffic resources remain AND queue not empty:
        choose oldest eligible request (alarm priority first)
        allocate one short transmission grant
        receive report and acknowledge sequence number
        remove acknowledged report from queue
    IF report is not acknowledged:
        retain it locally and retry with backoff
    release the resource immediately after the burst
    record active grants, blocked requests, delay and retries
```

The revised design exploits the fact that only a fraction of sensors are active at one time. It avoids holding a full-duration channel for an idle device. It introduces request signalling, queue state, scheduling and possible access collisions. Backoff, sequence numbers and acknowledgements are needed to avoid repeated collisions and duplicate updates.

D. Predict before implementing (1 mark)

Prediction: for a 10% duty-cycle workload, shared access should reduce average occupied resources from 100 reserved dedicated resources to approximately 10 active grants, while maintaining delivery

through queueing. Under synchronised bursts, the shared pool can still block if all devices request at once; queueing should convert some immediate blocking into delay rather than data loss.

E. Implement and validate (2 marks)

A simple counted comparison uses 100 devices, 10% average activity, 72 available traffic resources and the synchronised 100-device edge case. The dedicated approach reserves one resource per device in the design, although only about 10 are active on average. The shared approach allocates resources only to active requests. In the average case, approximately 10 resources are occupied and 62 of the 72-pool remain free. In the synchronised edge case, 100 requests exceed the 72-resource pool, so 28 cannot be served in the first scheduling instant; they remain queued for later grants.

Measure	**Dedicated allocation**	**Shared request/grant redesign**
Potential users	100	100
Average active devices at 10% duty	10	10 requests on average
Resources reserved/available	100 dedicated reservations (or 72 usable if capped)	72 shared resources
Average active resources	About 10	About 10
Average idle resources in a 72-resource pool	62 if only 10 are active	62; released immediately after bursts
Synchronous 100-request peak	28 blocked if only 72 usable	28 wait in queue; not discarded if queue is retained
Main risk	Wasted idle capacity	Queue delay and request contention

Validation Interpretation

The redesign does not create additional spectrum capacity. Instead, it improves the utilization of existing capacity when network activity is intermittent.

For the 100-request synchronous peak, the first scheduling interval still has a shortfall of 28 requests. However, a durable queue can serve these requests in subsequent intervals.

Actual delay and blocking must be measured using a time-based event simulator with packet arrival times and service durations. Therefore, the table represents a resource-count comparison rather than a measured end-to-end latency result.

F. Defend the Final Design (2 Marks)

The prediction is supported at the resource-count level because a shared pool avoids permanently reserving a channel for every silent device.

However, this approach introduces additional control signalling, scheduler complexity, queue memory requirements and delay during synchronised traffic peaks.

Dedicated FDMA/TDMA can remain preferable for continuous voice communication or strict low-latency traffic because it provides predictable resource access.

For low-rate campus telemetry, shared request/grant access with store-and-forward is the better choice.

Final Design Decision

Adopt shared scheduled access for routine telemetry, reserve a small priority resource for alarms, and retain dedicated resources only for continuous voice or strict Quality of Service (QoS) services.

This redesign addresses the identified idle-resource problem without assuming that a new spectrum allocation or a more advanced communication standard will automatically solve it.

Overall Conclusion

The seven questions demonstrate that multiple access is a resource-sharing problem involving time, frequency, code and spatial direction.

- FDMA and TDMA differ mainly in how they manage bursty traffic.
- CDMA depends on code correlation, synchronisation and power control.
- OFDMA involves balancing total throughput against fairness.
- SDMA depends on angular separation and antenna array resolution.

A practical wireless network combines multiple access techniques rather than relying on a single method for every deployment.

References and Assumptions

1. Primary source: ITA03-MP Assignment 2, Mobile Computing, CO2, pages 1–10 (user-provided assignment brief).

2. Standard engineering relations used:
    - Erlang and traffic-resource concepts.
    - Hadamard/Walsh orthogonality.
    - OFDM subcarrier orthogonality.
    - Uniform Linear Array (ULA) factor.
    - Jain's fairness index.
3. Numerical overhead percentages, channel-gain models, random seeds and simplified simulation assumptions are explicitly stated in the relevant sections.